

Hypothesis Testing (Part 2)

STAT 4530/6530
Statistical Inference for Data Scientists

Example: Climate Change

A news poll asked a random sample of 1008 Americans in 2019 whether they thought that climate change was a major threat to the stability of the United States.

Let p denote the population proportion who would say yes. To analyze whether this is a minority of the population ($p < 0.50$) or a majority of the population ($p > 0.50$), we test $H_0 : p = 0.50$ against $H_a : p \neq 0.50$.

Presuming that H_0 is true, the standard error of the sample proportion $\bar{y}$ is

$$se_0 = \sqrt{\frac{p_0(1 - p_0)}{n}} = \sqrt{\frac{0.50(1 - 0.50)}{1008}} = 0.0157.$$

In the poll, 542 people (accounting for 52% of the sample) said yes, so $\bar{y} = 0.520$ and the test statistic is

$$z = \frac{\bar{y} - p_0}{se_0} = \frac{0.52 - 0.50}{0.0157} = 1.26.$$

Example: Climate Change

The P -value is the probability that the Z test statistic equals the observed z value or a more extreme value that provides even stronger evidence against H_0 and in favor of H_a .

For $H_a : p \neq p_0$, the more extreme z values are the ones even further out in the two tails of the standard normal distribution, and the P -value is the two-tail probability that the absolute value of the Z test statistic is at least as large as the absolute value of the observed z .

In this example, $z = 1.26$, and the P -value is the probability that $Z \geq 1.26$ or $Z \leq -1.26$. The probability in one tail that $Z \geq 1.26$ is 0.104 and, by symmetry of the standard normal distribution about 0, the probability in the other tail is also 0.104. Thus the P -value is 0.208, which we will round to 0.21 (reporting it with an excessive number of decimal places makes it seem as if more accuracy exists than actually does).

One-Sided Significance Tests

A different alternative hypothesis predicts a deviation from H_0 in a particular direction. It has the form

$$H_a : p > p_0 \text{ or } H_a : p < p_0.$$

These hypotheses are called one-sided. In the climate change example, the P -value for either of the one-sided alternative hypotheses is 0.104.

In the gender-bias example, we had $H_0 : p = 0.40$ and $H_a : p > 0.40$, and the P -value was the probability of 9 or 10 out of 10 selections being male: $P(Y \geq 9) = 0.0017$.

- We needed a specific value for H_0 in order to know which value of p to use to compute the binomial probabilities.
- However, using a P -value of 0.0017 to conclude that it is implausible that $p = 0.40$ implicitly indicates that a broader null hypothesis of $H_0 : p \leq 0.40$ would also be implausible. Why?

One-Sided Significance Tests

In most research articles, significance tests use two-sided P -values.

- This partly reflects an objective approach to research that recognizes that an effect could go in either direction.
- In using two-sided P -values, researchers avoid the suspicion that they chose H_a when they saw the direction in which the data occurred, which is not ethical.
- Two-sided tests coincide with the usual approach in estimation.

Confidence intervals are two-sided, often obtained by adding and subtracting a margin of error from the point estimate, and there is strong connection between confidence intervals and two-sided tests, which we discuss later.

- One-sided confidence intervals are also possible, for instance having 95% confidence that a population proportion is no greater than 0.546 (i.e., between 0.0 and 0.546), but are not commonly used.

The Elements of a Significance Test for a Mean

For quantitative variables, significance tests usually refer to population means.

- The elements of a significance test for a mean resemble those for a proportion, except for using the t distribution instead of the standard normal distribution for the sampling distribution of the test statistic.

Assumptions: Significance tests for means assume that the data are obtained using randomization.

- The test we will present assumes a normal population distribution.
- This assumption is mainly relevant for small sample sizes and one-sided H_a , because otherwise the test is robust, performing adequately even when that assumption is violated.

The Elements of a Significance Test for a Mean

Hypotheses: The null hypothesis about a population mean μ has the form

$$H_0 : \mu = \mu_0$$

for a particular value μ_0 .

- This hypothesis usually refers to *no effect* or *no change* compared to some standard.
- For example, in a previous example we measured the population mean weight change μ for a therapy for anorexia, and the hypothesis that the therapy had no effect is $H_0 : \mu = 0$.

The most common alternative hypothesis is

$$H_a : \mu \neq \mu_0$$

- For the anorexia study, $H_a : \mu \neq 0$ states that the therapy has *some effect*.
- The one sided alternatives $H_a : \mu > \mu_0$ and $H_a : \mu < \mu_0$ predict a deviation in a certain direction from the H_0 value, and are less common.

The Elements of a Significance Test for a Mean

Test Statistic: In a study employing randomization, when the population distribution is normal, the sampling distribution of the sample mean $\bar{Y}$ is normal with a mean of μ .

- The evidence in the data about H_0 is summarized by the number of standard errors that the observation $\bar{y}$ falls from μ_0 .
- If the population standard deviation is σ , the standard error is $\sigma/\sqrt{n}$.
- As we previously did, we substitute the sample standard deviation s for σ and use an estimated standard error $se = s/\sqrt{n}$.
- The test statistic is

$$T = \frac{\bar{Y} - \mu_0}{S/\sqrt{n}}.$$

Under the null hypothesis, its sampling distribution is the t distribution with $df = n - 1$.

The Elements of a Significance Test for a Mean

P-value and Conclusion: The P -value is the probability, presuming H_0 , that the test statistic T equals the observed value t or a more extreme value that provides even stronger evidence against H_0 .

- For $H_a : \mu \neq \mu_0$, the P -value is the two-tail probability that T is at least as large in absolute value as the observed t , which is the probability that $\bar{Y}$ falls at least as far from μ_0 in either direction as its observed value.
- For one-sided alternatives, the P -value is a one-tail probability.
- As usual, the smaller the P -value, the more strongly the data contradict H_0 and support H_a .

Example: Political Ideology

On average, how politically conservative or liberal are U.S. citizens?

Political Ideology	Race/Ethnicity		
	Hispanic	Black	White
1. Extremely liberal	5	16	73
2. Liberal	49	52	209
3. Slightly liberal	46	42	190
4. Moderate, middle of road	155	182	705
5. Slightly conservative	50	43	260
6. Conservative	50	25	314
7. Extremely conservative	14	11	84
n	369	371	1835

- We can study political ideology by analyzing an item on the General Social Survey that has a seven-point scale of political views ranging from 1 (“extremely liberal”) to 7 (“extremely conservative”).
- The table above shows the scale and distribution of responses among levels for a recent GSS, with subjects classified by race/ethnicity.

Example: Political Ideology

Political ideology is on an ordinal scale. It is sometimes informative to treat ordinal data in a quantitative manner by assigning scores to the categories, in order to use the mean to summarize the responses.

- For the categories in this example, a mean below 4 shows a propensity toward liberalism and a mean above 4 shows a propensity toward conservatism.
- To test whether the data show much evidence of either of these, we conduct a significance test about how the population mean μ compares to the moderate value of 4.
- The hypotheses are $H_0 : \mu = 4.0$ and $H_a : \mu \neq 4.0$.
- H_0 states that, on average, the population response is politically “moderate, middle of the road.” H_a states that the mean falls in the liberal direction ($\mu < 4.0$) or in the conservative direction ($\mu > 4.0$).

Example: Political Ideology

We will illustrate the t test by conducting it for the Hispanic sample.

- The 369 Hispanic observations have sample mean $\bar{y} = 4.089$ and sample standard deviation $s = 1.339$.
- The estimated standard error is

$$se = \frac{s}{\sqrt{n}} = \frac{1.339}{\sqrt{369}} = 0.0697.$$

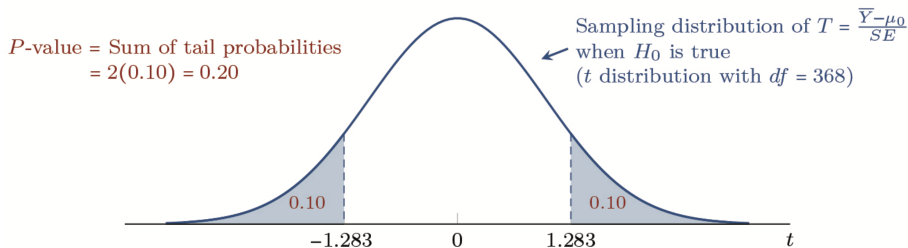
- The observed value of the test statistic is

$$t = \frac{\bar{y} - \mu_0}{se} = \frac{4.089 - 4.0}{0.0697} = 1.283,$$

with $df = 369 - 1 = 368$.

Example: Political Ideology

The observed value is $t = 1.283$, so the sample mean falls 1.283 estimated standard errors above the H_0 value.



The P -value is the two-tail probability, assuming H_0 is true, that T would exceed 1.283 in absolute value, which is 0.20. If the population mean ideology μ for Hispanics were 4.0, then the probability equals 0.20 that $\bar{Y}$ for 369 subjects would fall at least as far from 4.0 as the observed $\bar{y}$ of 4.089. In other words, the observed data would not be unusual. It is plausible that μ was 4.0, not in the conservative or liberal direction.

Example: Political Ideology

The t test assumes that the population distribution is normal, to ensure that the T test statistic has the t distribution.

- This seems inappropriate here, because the measurement of political ideology is discrete.
- As the sample size n increases in a study with randomization, however, this assumption becomes less important; an approximate normal sampling distribution occurs for $\bar{Y}$ (and a t null distribution for the T test statistic) regardless of the population distribution, by the central limit theorem.
- Two-sided inferences for a mean using the t distribution are robust against violations of the normal population assumption.
- The test does not perform well for a one-sided test when n is small and the population distribution is highly skewed. We will later discuss nonparametric tests that are better suited for that case.

Significance Tests Comparing Means

In practice, inferences about means or proportions usually involve comparisons for different groups.

- To compare the population means μ_1 and μ_2 of two groups, we test $H_0 : \mu_1 = \mu_2$.
- For the difference $\mu_1 - \mu_2$ parameter that we previously estimated with a confidence interval, the hypothesis is $H_0 : \mu_1 - \mu_2 = 0$, corresponding to *no effect* or *no difference*.
- A standard method assumes that the population standard deviations σ_1 and σ_2 for the two groups satisfy $\sigma_1 = \sigma_2$.
- Let Y_1 denote a randomly selected observation from group 1 and Y_2 a randomly selected observation from group 2. The hypotheses can be expressed as:

$H_0 :$ Both Y_1 and Y_2 have a $N(\mu, \sigma^2)$ distribution

$H_a :$ $Y_1 \sim N(\mu_1, \sigma^2)$, $Y_2 \sim N(\mu_1, \sigma^2)$ with $\mu_1 \neq \mu_2$.

Significance Tests Comparing Means

We estimate the common standard deviation σ by the pooled estimate s introduced earlier:

$$s = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{(n_1 - 1) + (n_2 - 1)}} = \sqrt{\frac{\sum_{i=1}^{n_1} (y_{i1} - \bar{y}_1)^2 + \sum_{i=1}^{n_2} (y_{i2} - \bar{y}_2)^2}{n_1 + n_2 - 2}}$$

- We do not expect the probability distributions for the groups to be *exactly* normal with *exactly* equal variances.
- In this case, however, two-sided tests are robust against violations of the assumptions of normality with common variance, especially when the sample sizes are similar and not extremely small.

Significance Tests Comparing Means

To test $H_0 : \mu_1 = \mu_2$ (equivalent to $H_0 : \mu_1 - \mu_2 = 0$), the test statistic is

$$T = \frac{(\bar{Y}_1 - \bar{Y}_2) - 0}{SE},$$

where the standard error SE is estimated by using the pooled standard deviation estimate s :

$$se = \sqrt{\frac{s^2}{n_1} + \frac{s^2}{n_2}} = s\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}.$$

Under H_0 , the T test statistic has a t distribution with $df = n_1 + n_2 - 2$.

Significance Tests Comparing Means

If the data show evidence of greatly different standard deviations (with, say, one being at least 50% larger than the other one), it is better to use an alternative t test that does not assume $\sigma_1 = \sigma_2$, called the **Welch t-test**.

- The Welch t -test uses the same form $T = (\bar{Y}_1 - \bar{Y}_2)/SE$ for the test statistic, but the standard error is estimated by

$$se = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}},$$

and df is replaced by a value designed so that T has a null distribution closely approximated by the t distribution.

- When approximately valid, we prefer the test that assumes $\sigma_1 = \sigma_2$, because this method relates better to modeling methods with additional variables (linear models), and because H_0 then more completely specifies “no effect.”
- In particular, in experimental studies with randomization, “no effect” naturally corresponds to entire distributions being identical, rather than merely the means.

Example: Comparing a Therapy to a Control Group

We previously used a confidence interval to compare mean weight changes in populations of anorexic girls receiving cognitive behavioral therapy or a control.

- We now test $H_0 : \mu_1 = \mu_2$ against $H_a : \mu_1 \neq \mu_2$.
- If the therapy truly has no effect relative to the control, the weight changes for the two populations would have equal means and equal standard deviations.
- The sample standard deviations are 7.31 and 7.99 for the cognitive behavioral therapy and control groups, respectively.
- The pooled standard deviation estimate is $s = 7.64$, and $\bar{y}_1 - \bar{y}_2 = 3.01 - (-0.45) = 3.46$ has $se = 2.06$.

Example: Comparing a Therapy to a Control Group

For testing H_0 , the observed value of the T test statistic is

$$t = \frac{\bar{y}_1 - \bar{y}_2}{se} = \frac{3.01 - (-0.45)}{2.06} = 1.68,$$

with $df = n_1 + n_2 - 2 = 29 + 26 - 2 = 53$.

- We get a P -value of 0.10, only weak evidence of different population means.

Example: Comparing a Therapy to a Control Group

This test can be performed in R:

```
> Anor <- read.table("http://stat4ds.rwth-aachen.de/data/Anorexia.dat", header=TRUE)
> cogbehav <- Anor$after[Anor$therapy=="cb"] - Anor$before[Anor$therapy=="cb"]
> control <- Anor$after[Anor$therapy=="c"] - Anor$before[Anor$therapy=="c"]
> t.test(cogbehav, control, var.equal=TRUE)
t = 1.676, df = 53, p-value = 0.09963
alternative hypothesis: true difference in means is not equal to 0

> t.test(cogbehav, control)
Welch Two Sample t-test # not assuming equal population standard deviations
t = 1.6677, df = 50.971, p-value = 0.1015
```

The output also shows the Welch t -test for comparing the means without assuming $\sigma_1 = \sigma_2$.

- It gives a similar result, with the same P -value (0.10) to two decimal places.

Effect Size for Comparison of Two Means

In this example, is the estimated difference of 3.46 between the mean weight gains for the two groups large, or small, in practical terms?

- As we will explain later, the P -value does not answer this question.
- A standardized way to describe $\bar{y}_1 - \bar{y}_2$ divides it by the pooled standard deviation estimate. This is called the **effect size**.
- We obtain the same effect size if we measure the weight gain in different units, such as kilograms or ounces.
- It is not sensitive to the sample sizes, unlike the standard error and P -value.

Effect Size for Comparison of Two Means

Whether a particular effect size is small, medium, or large depends on the substantive context, but an effect size of about 0.2 or less in absolute value is usually not practically important.

An effect size is considered to be large if the effect size is about 1, or larger, in absolute value.

With sample means of 3.01 and -0.45 pounds and a pooled standard deviation of $s = 7.64$ pounds, the effect size is:

$$\text{Effect Size} = \frac{\bar{y}_1 - \bar{y}_2}{s} = \frac{3.01 - (-0.45)}{7.64} = 0.45.$$

The difference between the sample means is a bit less than half a standard deviation, which is a moderate difference.

Significance Tests Comparing Proportions

To compare two groups on the population proportions p_1 and p_2 having a particular outcome, we test $H_0 : p_1 = p_2$ or, equivalently $H_0 : p_1 - p_2 = 0$.

- We assume independent samples of sizes n_1 and n_2 , with success counts Y_1 and Y_2 having binomial distributions and yielding sample proportion estimates $\hat{p}_1$ and $\hat{p}_2$.
- For studies employing randomization, the estimator $\hat{p}_1 - \hat{p}_2$ of $p_1 - p_2$ has an approximately normal sampling distribution, with variance

$$\text{Var}(\hat{p}_1 - \hat{p}_2) = \text{Var}(\hat{p}_1) + \text{Var}(\hat{p}_2) = \frac{p_1(1 - p_1)}{n_1} + \frac{p_2(1 - p_2)}{n_2}.$$

- The values p_1 and p_2 are not known, but under the presumption for H_0 that $p_1 = p_2$, we estimate their common value by a pooled estimate $\hat{p} = (y_1 + y_2)/(n_1 + n_2)$. The standard error is estimated by

$$\text{se}_0 = \sqrt{\frac{\hat{p}(1 - \hat{p})}{n_1} + \frac{\hat{p}(1 - \hat{p})}{n_2}} = \sqrt{\hat{p}(1 - \hat{p}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}.$$

Significance Tests Comparing Proportions

The test statistic measures the number of standard errors between the estimator $\hat{p}_1 - \hat{p}_2$ and the H_0 value of 0 for $p_1 - p_2$:

$$Z = \frac{(\hat{p}_1 - \hat{p}_2) - 0}{SE_0} = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1 - \hat{p}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

The P -value is a two-tail or one-tail probability from the standard normal distribution, depending on whether H_a is two-sided or one-sided.

Example: Prayer

Let us revisit the research investigation of the efficacy of prayer.

Group	Complications	No Complications	Total
Prayer	315	289	604
No Prayer	304	293	597

The proportion suffering complications after heart surgery was $\hat{p}_1 = 315/604 = 0.522$ for patients who had the prayer group and $\hat{p}_2 = 304/597 = 0.509$ for the patients not having a prayer group.

The pooled estimate of the H_0 value of $p_1 = p_2$ is $\hat{p} = (315 + 304)/(604 + 597) = 0.515$.

Reference

Agresti, A., & Kateri, M. (2021). Foundations of Statistics for Data Scientists: With R and Python (1st ed.). Chapman and Hall/CRC.